OpenMV® Super Telephoto Lens

Datasheet

SKU: Super-Telephoto-Lens-with-IR-Cut-Filter / Super-Telephoto-Lens

Description

This lens gives your OpenMV Cam the ability to see longer distances with extreme zoom. The standard lens has about a 70-degree FOV which allows your camera to see more things, but with less detail. This lens has about a 10-degree FOV, providing about a 9x zoom on any particular object. Available with or without a 650 nm IR cut filter.

Target Areas

Extreme long-range detection, eye tracking with IR, surveillance corridors, wildlife monitoring, traffic monitoring, industrial inspection at distance.

Key Features

- **~10-degree FOV (9x zoom)** — Up to 8.93x optical zoom with OV7725. Highest zoom in M12 lineup.
- **25 mm focal length** — F2.0 aperture, 1/3" image format.
- **M12 x 0.5 mount** — Standard M12 threaded mount compatible with all OpenMV Cam models.
- **Multi-sensor compatible** — FOV data for PS5520, PAG7936, MT9M114, MT9V024/034, OV7725, OV5640, and OV2640.
- **IR cut filter option** — Available with 650 nm IR cut filter or without.
- **3D model available** — STEP file available on GrabCAD.

Applications

The Super Telephoto Lens is great for any application where you need your OpenMV Cam to really zoom in on some subject. For example, the super telephoto lens makes it easy to do eye-tracking with IR light by zooming in on someone's eye, making it easier to determine the direction their pupil is pointing.

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1 Specifications

1.1 Optical Specifications

Component	Details
Focal Length	25 mm
Aperture	F2.0
Image Format	1/3"
Mount	M12 x 0.5 threaded
IR Cut Filter	650 nm (variant with IR cut filter)

1.2 FOV with PS5520 Sensor

Component	Details
Diagonal / Horizontal / Vertical	16.2° / 13.0° / 9.8°
Optical Zoom	6.94x (vs. 3.6 mm lens)

1.3 FOV with PAG7936 Sensor

Component	Details
Diagonal / Horizontal / Vertical	10.3° / 8.8° / 5.5°
Optical Zoom	8.93x (vs. 2.8 mm lens)

1.4 FOV with OV7725 Sensor

Component	Details
Diagonal / Horizontal / Vertical	11.0° / 8.8° / 6.6°
Optical Zoom	8.93x (vs. 2.8 mm lens)

1.5 FOV with OV5640 Sensor

Component	Details
Diagonal / Horizontal / Vertical	10.4° / 8.3° / 6.2°
Optical Zoom	8.93x (vs. 2.8 mm lens)

1.6 FOV with OV2640 Sensor

Component	Details
Diagonal / Horizontal / Vertical	10.1° / 8.1° / 6.0°
Optical Zoom	8.93x (vs. 2.8 mm lens)

1.7 FOV with MT9V024/034 Sensor

Component	Details
Diagonal / Horizontal / Vertical	12.2° / 10.3° / 6.6°
Optical Zoom	8.93x (vs. 2.8 mm lens)

1.8 FOV with MT9M114 Sensor

Component	Details
Diagonal / Horizontal / Vertical	7.1° / 5.6° / 4.2°
Optical Zoom	8.93x (vs. 2.8 mm lens)

2 General Info

Dimension	Value
Length	14 mm
Width	14 mm
Height	20 mm
Weight	5 g
Operating Temperature	-20°C to +80°C
Storage Temperature	-20°C to +80°C
Country of Origin	China
HS Code	8473.30.2000
UPC	602772227137, 638126314703
SKU	Super-Telephoto-Lens-with-IR-Cut-Filter, Super-Telephoto-Lens
ECCN	EAR99

3 Certifications

Certification	Region	Details
RoHS	EU	Compliant with EU Directive 2011/65/EU (RoHS 2). All components and solder are lead-free.
REACH	EU	Compliant with EU Regulation (EC) No 1907/2006.

4 Company Information

Field	Details
Website	https://openmv.io
Documentation	https://docs.openmv.io
Support	https://forums.openmv.io

5 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/super-telephoto-lens
3D Model (STEP)	https://grabcad.com/library/super-telephoto-lens-1

6 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.